

Chapter 2: Climate risk

We've seen that today's college students will experience around 2°C (3.6°F) of warming during their lifetime, unless more strenuous action is taken to reduce emissions of greenhouse gases. But I have not explained how this climate change affects humans and natural ecosystems that we care about. In this chapter, I will begin answering that question by introducing the concept of climate risk.

What is climate risk?

The IPCC [defines](#) risk in a somewhat elliptical way as “The potential for adverse consequences for human or ecological systems, recognising the diversity of values and objectives associated with such systems. In the context of climate change, risks can arise from potential impacts of climate change as well as human responses to climate change. Relevant adverse consequences include those on lives, livelihoods, health and well-being, economic, social and cultural assets and investments, infrastructure, services (including ecosystem services), ecosystems and species.”

Thus, the term *climate risk* refers to the range of challenges and potential harms that are created or intensified by climate change. This is based changes in the physical climate system (e.g., more intense rain), but it also reflects the extent to which these changes impact things we care about. A rain event by itself is not a climate risk, but one that falls on a city that is vulnerable to flooding is. In a future chapter, we'll talk about *transition risk*, which is the economic and social risks of transitioning to climate-safe technology.

Climate risk is therefore best understood as the combination of three interconnected components: *hazard, exposure, and vulnerability*:



Hazard refers to climate-related physical events that may cause loss of life, injury, or damage to property, infrastructure, livelihoods, service provision, or natural ecosystems. These hazards include sudden events like hurricanes, floods, and wildfires, as well as slow-moving processes such as sea-level rise, drought, and shifts in precipitation patterns. The frequency, intensity, duration, and spatial extent of these hazards are all influenced by climate change, with many becoming more severe and destructive as global temperatures rise.

Exposure represents the presence of things we care about (e.g., people, infrastructure, natural ecosystems) in places that could be adversely affected by climate hazards. For example, coastal cities have exposure to hurricanes and sea-level rise, while Western U.S. cities will be exposed to combination of extreme heat, wildfires, and drought. Most places have exposure to some type of climate risk.

Vulnerability is a measure of how easily an exposed system can be harmed by climate hazards, and this is heavily influenced by social and economic factors. For example, poorer communities have fewer financial resources to develop robust infrastructure (like storm drainage or building sturdier structures) and fewer options to evacuate or relocate, less insurance coverage, and reduced access to healthcare after disasters. They are also

often sited in worse places, like areas with poor drainage that richer communities don't want to build on.

This creates an unfair situation where two neighborhoods might face the same hazard (e.g., a hurricane) but the wealthier area is less vulnerable and therefore sustains less damage and recovers quickly while the poorer, high vulnerability neighborhood suffers lasting harm. Vulnerability, in other words, is mostly a social issue.

Climate risk emerges from the interaction of these three components, often conceptualized as $\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability}$. This multiplicative relationship emphasizes that risk factors combine and amplify each other. When all three elements intensify simultaneously, the resulting risk can increase exponentially rather than linearly.

It also creates situations where climate risk is *exposure limited*. If population in some region is very low, so exposure is small, then even if the hazards are getting significantly worse, it won't represent significant climate risk.

This formula also highlights that interventions addressing any single component can significantly reduce overall climate risk. Communities may have limited ability to directly reduce hazards in the short term but can effectively lower risk by reducing exposure and vulnerability through zoning restrictions or improved building standards, early warning systems, and social safety nets.

Let's now examine each term individually.

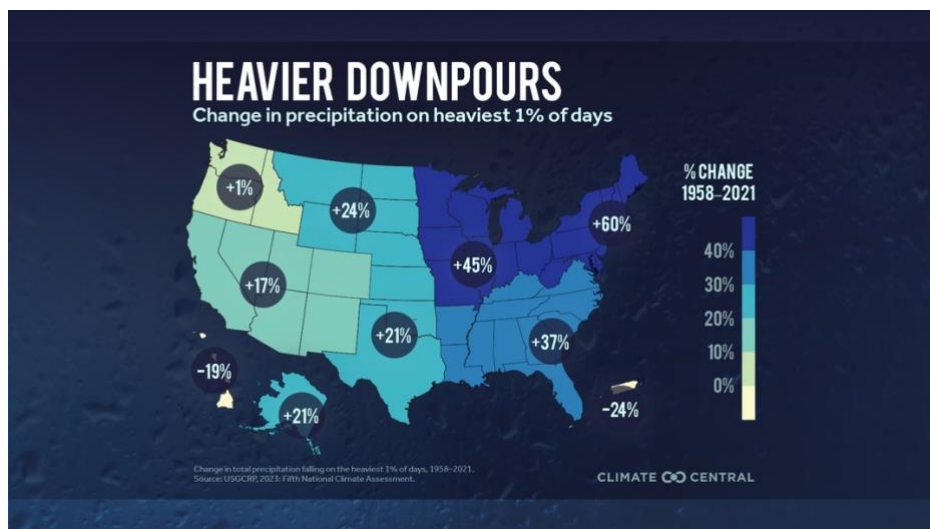
Hazards

Temperature: With 100% confidence, we know that the Earth's average temperature is rising. This is a serious risk, as we'll talk about later, we are adapted to our present climate.

Precipitation: When it comes to precipitation, the intensity of the rainfall, the frequency of occurrence, and the duration of the rainfall all matter for climate risk. This makes analyzing precipitation more complicated than analyzing temperature.

In general, warming increases the **intensity** of precipitation. The reason is straightforward physics: warmer air can hold more water vapor: the atmosphere's water-holding capacity increases by about 7% per °C increase in local air temperature. If the air converging into a storm remains the same, then the water vapor flowing into a storm increases as the climate warms. As the air ascends and cools in the storm's updraft, that extra water vapor condenses, producing more intense rainfall. In other words, with the same dynamics but a warmer background climate, storms wring out larger amounts of precipitation.

The net result of this is that the heaviest downpours are growing more intense. For example, here is a plot of how the amount of rain in the heaviest 1% rain days has increased:



Plot from [Climate Central](#), based on Fifth National Climate Assessment. Retrieved 2025-09-13.

These heaviest downpours are of great concern for climate risk. In an intense storm, soil can only absorb water so fast, so much of the rain runs off instead of soaking in. This runoff can overwhelm both natural drainage and built infrastructure, causing flooding.

Because intensity of the heaviest events is increasing, the frequency of rain will decline. Put differently, when it rains, it will rain harder, and when it's dry, it will be dry for a longer time. Combined with warmer temperatures — which increase the rate at which soils lose water through evaporation — this increases the likelihood of drought.

The result is what scientists call [*hydrologic whiplash*](#): both wet and dry extremes become more common. Wet extremes bring flooding, erosion, and landslides. Dry extremes bring water shortages, stressed ecosystems, and crop losses. Dry periods also intensify heat, creating feedbacks that link these hazards together.

There are several other important changes in precipitation. Although global total rainfall is expected to increase, the increase will not be distributed evenly. Some places (particularly the subtropics) will see long-term declines of rainfall. In other places, the local increase in the rate of evaporation (due to warmer temperatures) will outweigh the local increase in precipitation and such places will dry out even as rainfall increases.

There will also be shifts in the form of precipitation. Less wintertime precipitation will fall as snow and more will fall as rain. This is more important than it might sound: When snow falls in winter, the water does not run off until the snow melts in spring. Rain, on the other hand, runs off immediately, so changing the form of precipitation changes the timing of runoff, increasing the availability of water in winter and early spring and decreasing it in late spring and summer. Given that humans have adapted to the present timing of runoff, changes will negatively impact human welfare.

Floods are connected to precipitation. Given the increase in extreme precipitation, we would expect floods to be getting worse, but it's more complicated than that. The reason is that floods are also affected by other human activities (e.g., development, dams, levees), which can offset the impacts of more extreme rainfall. Observations therefore show no clear worldwide signal, though certain regions have experienced increases tied to both climate change and development choices. Despite the lack of a trend, our confidence in how precipitation will change gives us good reasons to believe that floods will become more severe as the climate warms unless humans work very hard to adapt to heavier rain events.

Sea-level rise is a 100%-certain impact of climate change for two reasons. First, as grounded ice melts, the meltwater runs into the ocean, increasing the total amount of water in the ocean and, therefore, sea level. Second, like most things, water expands when it warms, which also tends to raise sea level. Measurements confirm that sea levels have been rising as temperatures have gone up, and we can be certain that the seas will continue to rise for a very long time into the future.

The latest IPCC report predicts that sea level will rise 47 to 73 cm (19 to 29 inches) above 1995-2014 levels by 2100. While this might sound manageable, the danger lies in how it amplifies extreme sea-level events. Think of sea level rise as raising the baseline water level upon which storm surges, king tides, and other storm-driven extreme sea level events ride. [[[update with new IPCC]]]

A coastal defense system designed to handle a 5-meter extreme sea level event works fine when a 4.8-m storm surge comes along, but it is inadequate when that same surge occurs on top of a 0.5-meter higher sea level. In other words, sea level rise can push flooding beyond critical thresholds, overwhelming drainage systems, breaching seawalls, and inundating areas previously considered safe.

Over thousands of years, sea level rise will be very large. Paleoclimate data tells us that we can (roughly) expect a several meters (tens of ft) for the first 2°C of warming. Beyond that, each degree will be responsible for several tens of meters (~100 ft) of sea-level rise. The only good news is that these huge increases will take a very long time. Thus, the 1.3°C of warming we have already experienced means that we are committed to something in the neighborhood of 6-13 m (20-40 feet) of sea-level rise over the next few thousand years. This will irreversibly alter the shoreline of the planet, for a very long time.

Ocean acidification is another 100% certain consequence of emissions of carbon dioxide to the atmosphere. This is the result of a simple chain of logic: 1) humans are dumping carbon dioxide into the atmosphere, 2) a significant fraction of this carbon dioxide is absorbed by the oceans where, in the liquid environment of the ocean, carbon dioxide is converted into carbonic acid.

Thus, as humans emit carbon dioxide, the oceans become more acidic. There's no mystery or uncertainty in this process—we can observe that today's surface waters are [more acidic](#) than they have been in the last two million years. This trend will persist as long as humans continue burning fossil fuels.

This can greatly affect ocean ecosystems. Making oceans more acidic reduces carbonate ions, which makes it harder for animals with shells and skeletons to build and maintain them. Eventually, the increased acidity can become deadly for them. While the acidity of the ocean has varied over geologic times and ecosystems have adjusted, the changes occurring now are extremely fast, so it is far from certain that natural evolutionary processes can buffer the impacts.

Tropical cyclones are called hurricanes if they form in the Atlantic, typhoons in the Northwest Pacific, and cyclones in Southwest Pacific or Indian Ocean. They are among the

most expensive and destructive weather events humans deal with. I'll refer to them in this section as tropical cyclones, abbreviated TCs.

Let me explain why we are 100% certain that changes in the climate system are increasing the hazard from TCs:

1. Sea level

The most direct and certain way that climate change amplifies tropical cyclone hazards is through coastal flooding. Human-caused global warming is driving sea level rise, which continuously elevates the baseline water level from which storm surges operate.

Consequently, there is **high confidence** that coastal storm surges caused by TCs will increase globally in the future, magnifying coastal flooding risks even if storm winds themselves did not change

2. Rainfall

As discussed above, thermodynamics dictates that a warmer atmosphere holds more moisture. This means that TCs occurring in a warmer climate should see higher rain rates. There is already **medium confidence** that the rain intensity of TCs have increased globally with recent warming. Looking to the future, there is **high confidence** that TC-associated precipitation (both mean and extreme rates) will intensify across all warming levels.

3. Intensity

Based on observations, we have **medium confidence** that the global portion of intense Category 3–5 storms increased over the past several decades. In fact, in most ocean basins, the absolute number of the most severe TCs (category 4 and 5) has shown a documented increase.

However, attributing these intensity trends to climate change is complex. The underlying greenhouse warming signal is complicated and, in some regions, temporarily masked by anthropogenic aerosol emissions combined with natural variability. For example, a cooling

trend in the East Pacific combined with rapid Atlantic warming has driven a robust increase in North Atlantic TC frequency (**medium confidence**) while simultaneously driving a robust decrease in the Western North Pacific (**high confidence**).

Looking forward, there is **high confidence** that both global TC wind intensities and the proportion of Category 4 and 5 storms will continue to rise.

4. Number of tropical cyclones

While models consistently project that TCs that do form will be more severe, there is **low confidence** in projections of how the *total number* of TCs will evolve into the future. The reason is that the scientific community lacks a comprehensive theory explaining what determines the annual global frequency of TCs. Each year, approximately 80 of these storms form across the world's oceans, but we don't know why nature settles on this number rather than 8 or 800.

This theoretical gap means our projections of TC numbers rely almost entirely on computer simulations of the climate, which struggle to reliably simulate small-scale, complex phenomena like TC formation. This should give us pause when considering their forecasts about future TC frequency. Regardless of how the number of storms change, though, please remember that, when TCs do form, they will tend to be stronger, rainier, and riding on a higher sea, making them more hazardous than TCs that are forming today.

This discussion has only touched on a few of the ways that TCs are getting more hazardous. There are other possible changes, such as the tracks of TCs moving to higher latitudes, more frequent rapid intensification, or the slowdown of TC translation speed, all of which can also increase TC hazard.

Wildfires are another extremely serious impact of climate change. These fires are a natural part of many ecosystems, but human-caused climate change is altering their character in important ways.

When talking about wildfires, it's useful to start with **fire weather** — the compound atmospheric conditions of high temperatures, low relative humidity, low soil moisture, and strong winds that promote wildfires. The dry atmosphere desiccates the fuel (trees, grass), priming it to burn rapidly. The strong winds ensure that any fire that starts will spread rapidly.

Extreme fire weather has increased globally (**high confidence**). This trend is primarily driven by rising temperatures due to human-caused climate change (**high confidence**). While greenhouse gases have driven this drying, aerosol-forced cooling during the twentieth century temporarily counteracted and masked this trend.

Satellite records show a decline in total global burned area (**medium confidence**) primarily driven by a large decline in savanna and grassland fires in sub-Saharan Africa due to landscape fragmentation, agricultural conversion, and land management. In contrast, forest loss due to fire have increased (**medium confidence**), driven by rising fire activity in extratropical and carbon-rich boreal forests (**high confidence**). Simultaneously, extreme wildfires, megafires, and wildfire disasters have been increasing over recent decades.

Wildfire dynamics are also shifting: wildfire seasons are lengthening globally due to warmer, drier seasons and earlier snowmelt; nighttime fire intensity is rising as nights warm and dry; and spatially concurrent fire weather extremes are increasingly co-occurring across multiple regions (**medium confidence**).

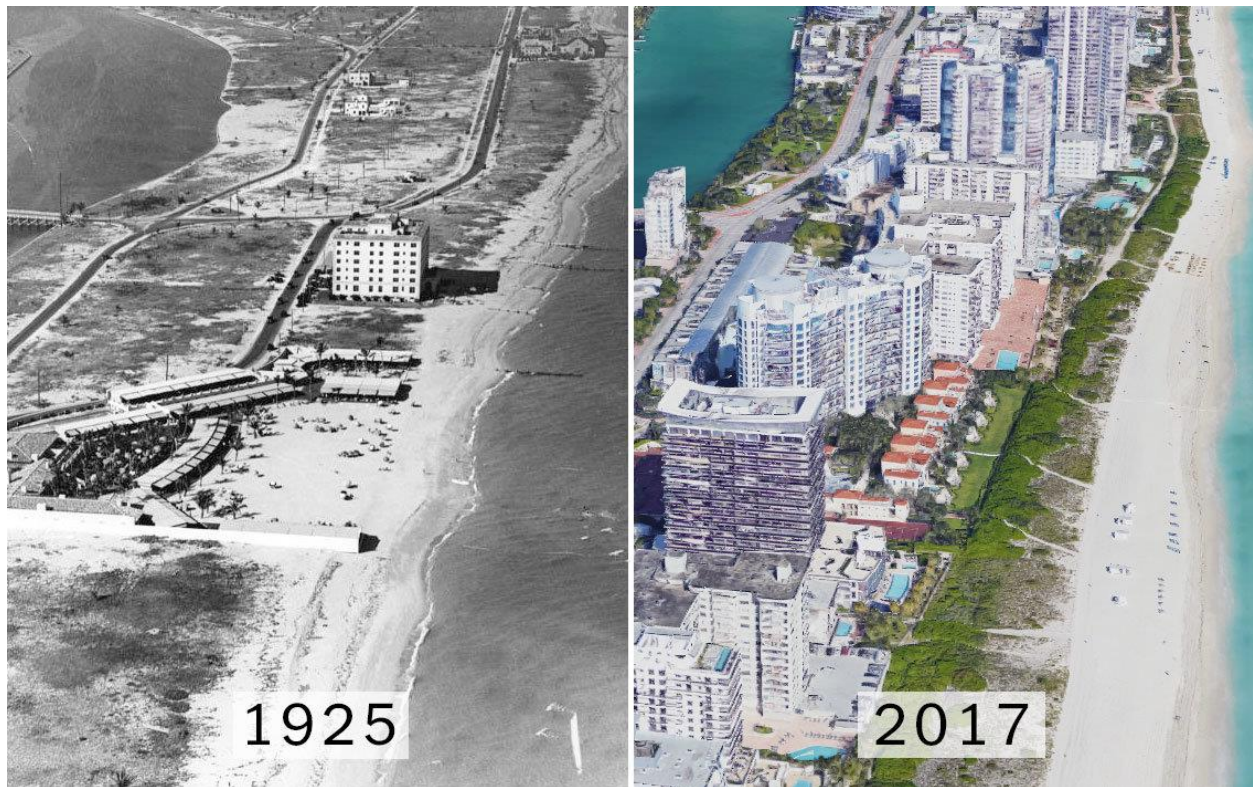
Projections indicate widespread increases in extreme fire weather and season length with continued global warming (**high confidence**). Under continued warming, megawildfires will become more frequent and widespread (**high confidence**).

Wildfire smoke is another hazard. It can spread thousands of miles, reducing air quality and causing respiratory and cardiovascular health problems. In many regions, smoke exposure already represents one of the largest climate-related health burdens, [killing 24,000 people per year](#) in the U.S.

Exposure

After hazard, the next factor in climate risk is exposure. Over the past century, humanity has dramatically increased its exposure to climate-related hazards through population growth and increasing wealth, leading to massive infrastructure development in vulnerable locations. This expansion into hazardous regions represents the biggest driver of our growing climate risk in many regions.

Coastal zones illustrate this problem most vividly. The allure of oceanfront property, maritime commerce, and tourism has driven a concentration of people and infrastructure on the coast, precisely where severe storms and sea level rise pose significant threats. For example, here is a photo of Miami Beach from early in the 20th and 21st centuries:



<https://static01.nyt.com/images/2017/09/08/climate/08HURRICANERISK-miami-beach/miami-beach-superJumbo.jpg>

This shows the enormous build-out in areas that are susceptible to many different types of hazard.

Similarly concerning is the expansion of housing into what ecologists call the "wildland-urban interface" (WUI)—the transition zone between wilderness and human development. Throughout the American West, Australia, Mediterranean Europe, and other fire-prone regions, residential development has pushed deeper into forested areas that naturally experience periodic wildfires.

The attraction of these woodland settings is understandable. They offer privacy, natural beauty, and escape from urban congestion. However, this migration pattern has effectively moved millions of people directly into the bullseye of the worsening climate hazard of wildfire, causing exposure to skyrocket.

Vulnerability

Vulnerability is a measure of the susceptibility of communities, infrastructure, and other human and natural systems to be damaged when exposed to climate hazards. While hurricanes, floods, and wildfires might seem like equal-opportunity disasters, their impacts are dramatically uneven, often following existing social inequalities. The most vulnerable segments of society — the elderly, the poor, those with chronic health conditions, and marginalized communities — frequently bear a disproportionate burden of climate impacts.

As an example of vulnerability, here is a photo of a city in Florida after a hurricane came ashore:



Most of the houses were wiped off their foundations, but the house in the foreground is basically unaffected. From the [NY Times article](#) that the photo came from:

Dr. Lackey said he and Mr. King, who jointly own the Mexico Beach house, did not even refer to the minimum wind resistance required in Bay County. They built the sand palace to withstand 250 mile-an-hour winds.

The house was fashioned from poured concrete, reinforced by steel cables and rebar, with additional concrete bolstering the corners of the house. The space under the roof was minimized so that wind could not sneak in underneath and lift it off. The home's elevation, on high pilings, was meant to keep it above the surge of seawater that usually accompanies powerful hurricanes.

This demonstrates one of the key truths in climate risk: With enough money, you can engineer yourself out of most climate impacts.

Hurricane Katrina's devastating impacts on New Orleans' lower-income neighborhoods compared to wealthier areas shows this clearly. The Lower Ninth Ward, a predominantly Black, lower-income community, suffered catastrophic flooding, with water depths reaching 12 feet in some areas, destroying nearly every structure. Many of the residents who perished in the storm lacked the means to evacuate (e.g., private vehicles, limited financial resources) and had nowhere to go even if they could evacuate. Those who survived often had inadequate insurance coverage to fund recovery.

By contrast, wealthier neighborhoods like the French Quarter experienced significantly less flooding due to their higher elevation, stronger housing construction, and better flood infrastructure. Their residents typically had greater access to transportation, financial resources to evacuate preemptively, and robust insurance policies to facilitate recovery.

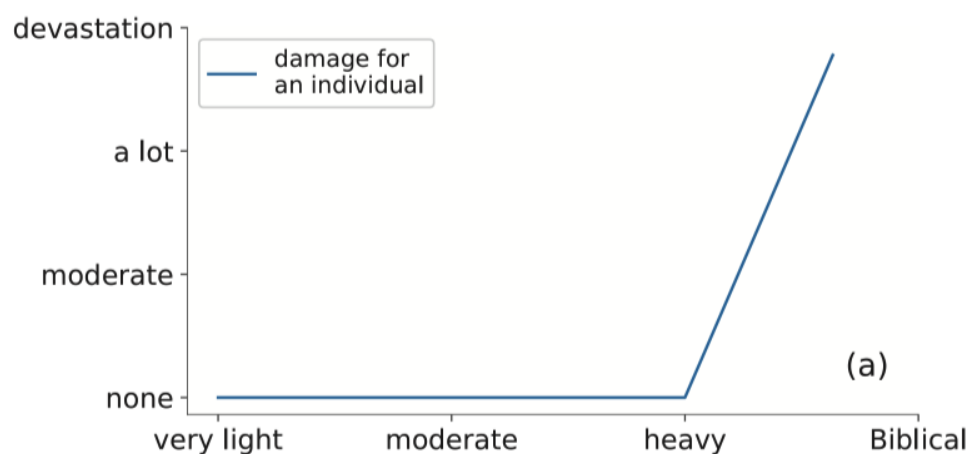
The disparity extended into the recovery phase: several years after Katrina, affluent areas had largely been rebuilt, while the Lower Ninth Ward still contained vacant lots and abandoned buildings. This pattern exemplifies how socioeconomic factors — including income, race, housing quality, transportation access, and political influence — transform the same hazard and exposure into vastly different lived experiences, with these differences persisting for generations.

This uneven distribution of climate impacts across society reveals something profound about our climate crisis: *it's not merely an environmental challenge but a complex social problem where social structures determine who suffers and who does not.*

Non-linearity

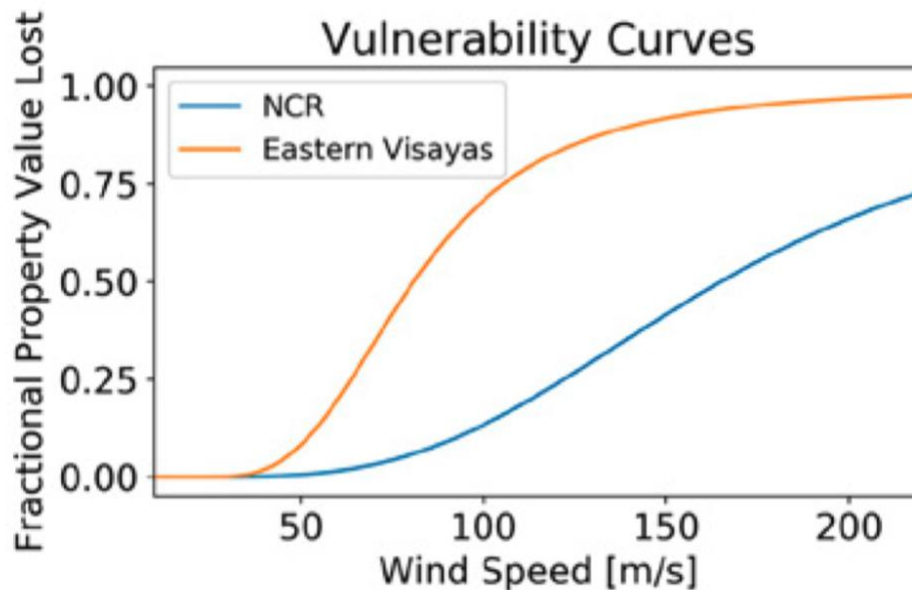
Beyond wealth, another crucial factor shaping vulnerability is non-linearity. If climate impacts were linear, each additional increment of warming — say, another 0.1°C — would produce the same increment of damage. Things would slowly get worse, and over several decades, the accumulated damages would become enormous. This linear model has dominated much of our thinking about climate change: a gradual, predictable process where impacts scale proportionally with warming.

But that's not how it works. Rather, things are fine until they're not. For example, this figure shows a schematic of how increasing rainfall can impact an individual:



At low and moderate rates, rainfall does no damage because existing infrastructure is built to handle it. The point where damage begins (beyond “heavy” rainfall) is a threshold, and it is largely an engineered one. Society has chosen how much rainfall its drainage systems, levees, and buildings can handle, often based on the climate of the past. Beyond that threshold, damage rises rapidly: You go from zero damage if floodwater stops half an inch below your front door to tens of thousands of dollars if the water rises one more inch and flows inside.

Here are realistic [vulnerability curves](#) for extreme winds for two regions in the Philippines:

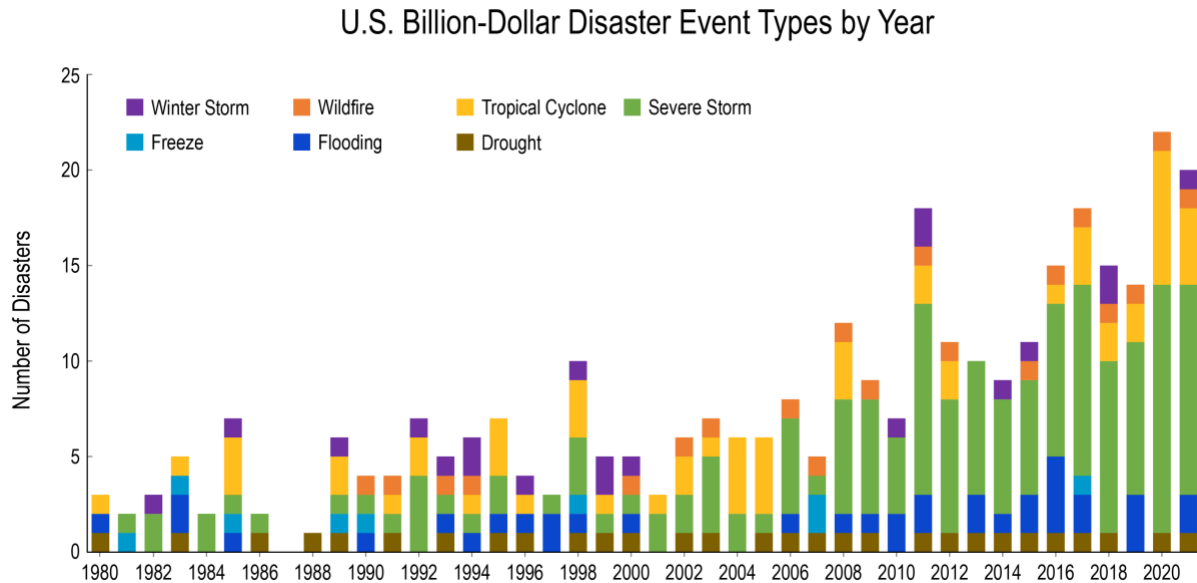


The National Capital Region (NCR) contains Manila and is the most populated, developed, and urbanized area in the Philippines. It has strong infrastructure and high socioeconomic resilience. Eastern Visayas is a poorer, predominantly rural region characterized by weaker building construction, greater vulnerability to high winds, and significantly lower capacity to cope with disaster-related losses. As a result, at every wind speed, the richer region (NCR) shows less damage.

The plot also shows the non-linearity of vulnerability: zero damage at low wind speeds, then rapidly increasing when wind speeds pass a threshold. The thresholds are set at least partially by wealth, so the poorer region (Eastern Visayas) sees damages start to rise rapidly at a lower wind speed than the richer region (NCR).

Climate change and impacts

You may have seen a graphic of billion-dollar disasters in the U.S. over time:



From: <https://ncics.org/cics-news/billion-dollar-disasters-are-happening-more-often/>

Given that climate risk is the combination of these three factors: hazard, exposure, and vulnerability, we can decompose the trend in billion dollar disasters into trends in these underlying factors.

We know that most of this trend is driven by exposure: more people, who are richer, living in places where hazards hit (e.g., near the water, in the wildland-urban interface).

We can also be confident that climate hazards are increasing in frequency and intensity and this is also contributing positively to the trend.

At the same time, vulnerability is declining. For example, after Hurricane Andrew devastated South Florida in 1992, the state adopted hurricane safety standards that mandated more resilient construction. This included requiring impact-resistant windows or shutters, banning particle board in coastal areas, and reinforcing roofs and walls with stronger connections and concrete supports. Miami-Dade and Broward counties were designated as High-Velocity Hurricane Zones, with even stricter wind-load requirements.

This building code update works. Research after subsequent hurricanes has shown that homes built under the post-Andrew code sustained significantly less damage than older structures.

In the aftermath of Hurricane Harvey in 2017, Houston and Harris County moved to tighten flood-related building regulations: they shifted from using the 100-year floodplain to the 500-year standard and required new structures in those zones to be built at least two feet above that. Alongside code changes, a \$2.5 billion bond referendum passed in 2018 funded flood control projects such as detention basins and bayou improvements.

There are limits to vulnerability reductions, however. For example, after Hurricane Ike made a near-miss on Houston in 2008, Texas began planning for a seawall to protect Galveston Bay and the Houston Ship Channel. Such a structure would be extremely expensive (\$30-\$60B) and no one seems willing to pay for it. So the project has been in a holding pattern for nearly two decades.

Putting this all together, the rising toll of climate-related disasters can be clearly understood through the changes in its core components. Climate change is intensifying the frequency and severity of hazards. Exposure is also increasing as people move to high-risk areas. Counteracting these trends is reduced vulnerability, which cancels some of the increases in hazards and exposure.